

Week 2 — Variables & Math

Name: ____ Date: ____

A **variable** is a labeled box in the computer's memory. = puts a value in the box — and the **Workspace pane** shows every box, live.

Today's words

Word	What it means
variable	a labeled box that holds one value
Workspace	the pane showing every box — the computer's memory, live
assignment	putting a value in a box: <code>score = 10</code>
reassignment	replacing what's in the box (<code>score = score + 10</code> — take, add, put back!)
semicolon ;	the hush key — the line still works, just silently
ans	MATLAB's scratch box for unnamed answers
operator	a math symbol: <code>+</code> <code>-</code> <code>*</code> <code>/</code> <code>^</code> (plus the machines <code>floor()</code> and <code>mod()</code>)

1 • Be the computer 🤖

Follow the code *exactly* and write what the screen shows. (Careful — some lines are hushed, and the last one is not!)

```
x = 10;  
disp(x)  
disp('x')  
x = x + 5
```

Line 1 shows: _ **Line 2 shows:** _ Line 3 shows: ____
Line 4 has no hush key, so MATLAB announces: `x =` ____
After all four lines, the Workspace row for `x` holds: __

2 • PEMDAS drill — evaluate like MATLAB 🧮

```
disp(2 + 3 * 4)      % → ____  
disp((2 + 3) * 4)    % → ____  
disp(10 - 2 * 3)     % → ____  
disp(2 ^ 3)          % → ____
```

3 • The pizza machines 🍕

23 slices arrive for 4 kids.

```
disp(floor(23 / 4))  % slices EACH: ____  
disp(mod(23, 4))    % slices LEFT OVER: ____
```

Check yourself: (each × 4) + left over should equal __.

🧠 Brain teaser (optional — take it home)

The two-box swap. Real programmers get asked this in job interviews!

Box `a` holds **3**. Box `b` holds **7**. Make them trade values — `a` ends with 7 and `b` ends with 3. Rules:

- Every step must be an assignment (something like `box = box`).
- A box holds only ONE value — putting something in destroys what was there.
- You can't "remember" a number in your head. Boxes only!

Hint: `a = b` first destroys the 3 forever... so maybe you need a third box? Write your steps like code. Bring it next week for a shout-out.